LiquidCrystal()

The `LiquidCrystal()` function creates a new instance for the library (so we can use the library in our sketches), gives it a variable name, and assigns the pin numbers for the LCD. This function must be used the first time we want to use the library. There are several different formats for the functions syntax, but we will stick with the following simple one:

```
LiquidCrystal name(rs, en, db4, db5, db6, db7)
```

In this function, `name` can be any unique name that we want following the normal rules for variable names, but to make things easier we will stick with the convention of using `lcd` in our code. To make wiring easier and to leave certain I/O pins open on the Arduino board, we will skirt tradition here and assign the digital pins 5, 6, 7, 8, 9, 10 as the pin numbers for the rs, en, db4, db5, db6, db7 pins respectively. This comes together in our source code as the following line:

```
LiquidCrystal lcd(5, 6, 7, 8, 9, 10);
```

Now that we have created an instance of the LiquidCrystal library, we will use the variable name `lcd` in front of each of the functions that are a part of that library, as you will see in the next function.

begin()

With the library instance created, we need to start the library and define what size LCD we intend to use. To do that, we use the function `begin()` inside the `setup()` function.

```
begin(cols, rows)
```

For a 16-character wide by 2-character tall LCD like the one we have been using, we would use `16, 2` for the function parameters. For larger or smaller sized displays we would adjust these numbers to suit, as in `20, 4` or `8, 1`. The function as it appeared in our example follows:

```
lcd.begin(16, 2);
```

This statement links the `begin()` function to our instance `lcd` and specifies a 16×2 display. Now that we've got the library set up correctly, let's look at some of the functions that we can use with this library.

■ **Note** While LCDs of other sizes will work for these examples, the code may act a little odd or need modification.

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print()

To actually send information to our LCD, we can use the `print()` function for basic character strings in one of the following two syntax formats:

```
print(data)
print(data, BASE)
```